

```
Terminal web_text.py x
1 # Assignment 1, Page 2: Access Text from the Web
2 from urllib.request import urlopen
3 from bs4 import BeautifulSoup
4 from nltk.tokenize import word_tokenize
5
6 extinction_url = "https://www.bbc.com/news/science-environment-61242789"
7 extinction_html = urlopen(extinction_url)
8 extinction_html_parse = BeautifulSoup(extinction_html, 'html.parser')
9
10 for index, element in enumerate(extinction_html_parse.find_all('p')):
11     words = element.get_text()
12     print(f'\nTokens in Paragraph {index + 1}: {word_tokenize(words)}')
13
14 for index, element in enumerate(extinction_html_parse.find_all('b')):
15     words = element.get_text()
16     print(f'\nTokens in Bolded Text: {index + 1}: {word_tokenize(words)}')
17
18 from nltk.tokenize import word_tokenize
19
20 with open('exercises/local_article.txt') as local_text_file:
21     raw_local_text_file = local_text_file.read()
22     print(f'\nTokens from the Local Article Called 'local_article.txt': {word_tokenize(raw_local_text_file)}')
23
24 from nltk.tokenize import word_tokenize
25
26 user_input = input('Enter Some Random Text: ')
27 print(f'You typed {len(word_tokenize(user_input))} words.')
```

Access Text from the Web

Extract Text from the Web

NLP is not only limited to analyzing text found in a pre-existing corpus. We can scrap text from a website and then process it. In this example, we are going to use an article from the BBC. Before we can start extracting text from the web, we need to first install the BeautifulSoup package.

```
python3 -m pip install beautifulsoup4==4.11.1
```

We are ready to begin coding. Open the `web_text.py` file with the button below.

Open `web_text.py`

Import the processes we will use to extract text in the exercises on this page.

- Using the `urllib.request` package, we can import `urlopen`.
- Using the `bs4` package, we can import `BeautifulSoup`.
- Using the `nltk.tokenize` package, we can import `word_tokenize`.

```
from urllib.request import urlopen
from bs4 import BeautifulSoup
from nltk.tokenize import word_tokenize
```

Create some variables used for the various stages of parse text from a website.

- We first feed the computer the specified URL, which we called `extinction_url`.
- Second, we used the `urlopen` method to retrieve the raw text from the variable's URL page, then store its value in the variable called `extinction_html`.
- Finally, we created a third variable called `extinction_html_parse` to use the `BeautifulSoup` on the variable called `extinction_html`.

```
extinction_url = "https://www.bbc.com/news/science-environment-61242789"
extinction_html = urlopen(extinction_url)
extinction_html_parse = BeautifulSoup(extinction_html, 'html.parser')
```

Extracting text is done by HTML tag. We are going to use the `qp` tag along with the `find_all` and `get_text` methods. This will return the text from all of the paragraphs on the website.

Enter the code below, then select the `TRY IT` button to run the file.

```
for index, element in enumerate(extinction_html_parse.find_all('p')):
    words = element.get_text()
    print(f'\nTokens in Paragraph {index + 1}: {word_tokenize(words)}')
```

TRY IT

Command was successfully executed.

Articles on the internet can change over time, so your output may differ from the one shown below. Your output should be, for the most part, similar. For brevity's sake, not all of the paragraphs are show:

```
Tokens in Paragraph 1: ['By', 'Helen', 'BriggsEnvironment', 'correspondent']
Tokens in Paragraph 2: ['One', 'is', 'five', 'reptiles', 'is', 'threatened', 'with']
Tokens in Paragraph 3: ['Scientists', 'are', 'calling', 'for', 'urgent', 'conserved']
Tokens in Paragraph 4: ['They', 'say', 'reptiles', 'have', 'long', 'been', 'overlooked']
Tokens in Paragraph 5: ['So', 'far', ',', '31', 'species', 'have', 'gone', 'extinct']
Tokens in Paragraph 6: ['The', 'study', ',', 'published', 'in', 'Nature', ',', 'too']
```

Try this variation:

Write the code to produce the following expected output, then select the `TRY IT` button to run the file and check your work.

The expected output is shown below.

```
Tokens in Bolded Text: 1: ['One', 'is', 'five', 'reptiles', 'is', '']
```

TRY IT

Command was successfully executed.

▼ Solution

One possible solution is shown below, where we can use a `for` loop along with `find_all` and `get_text` to return all words in the `p` tags.

```
for index, element in enumerate(extinction_html_parse.find_all('p')):
    words = element.get_text()
    print(f'\nTokens in Bolded Text: {index + 1}: {word_tokenize(word
```

Extract Text from Local Files

We can also apply the same NLP concepts to text files on our local computer. The `local_article.txt` file resides in the `exercises` directory. Be sure to import the `word_tokenize` function so we can tokenize the text.

Enter the code below, then select the `TRY IT` button to run the file.

```
word_tokenize

local_article.txt' as local_text_file:
    local_text_file.read()
    # Local Article Called 'local_article.txt': {word_tokenize(raw_local_text_file)}")
```

TRY IT

Command was successfully executed.

Your output should display as shown below:

```
Tokens from the Local Article Called 'local_article.txt': ['One', 'is', 'five', 're
```

Try this variation:

We can also extract tokens from user input.

Enter the below code, then select the `TRY IT` button to run the file. Codio will open the terminal where you can type several words (separated by spaces). Press `ENTER` when done.

```
from nltk.tokenize import word_tokenize

user_input = input('Enter Some Random Text: ')
print(f'You typed {len(word_tokenize(user_input))} words.')
```

TRY IT

Your output will look something like this:

```
Enter Some Random Text: cat dog meow
You typed 3 words.
```

Reading Question

Fill in the blanks.

Statement	Solution
We use <code>urlopen</code> to extract HTML raw text from URLs.	<code>urlopen()</code>
We use <code>urlopen</code> to extract raw text from local files.	<code>open()</code>

- We use `urlopen()` to extract HTML raw text from URLs.
- This is imported from the URL Request Library.
- We use `open()` to extract raw text from local files.

CHOOSE AN ANSWER

Next